Trend Rotation and Multi-Asset Momentum

Two systematic strategies compared: SPY/IEF moving-average rotation and a six-ETF monthly momentum rotation

Prepared October 19, 2026

Hypothetical backtested results. For discussion purposes only. Generated by an AI multi-agent research pipeline.

Executive Summary

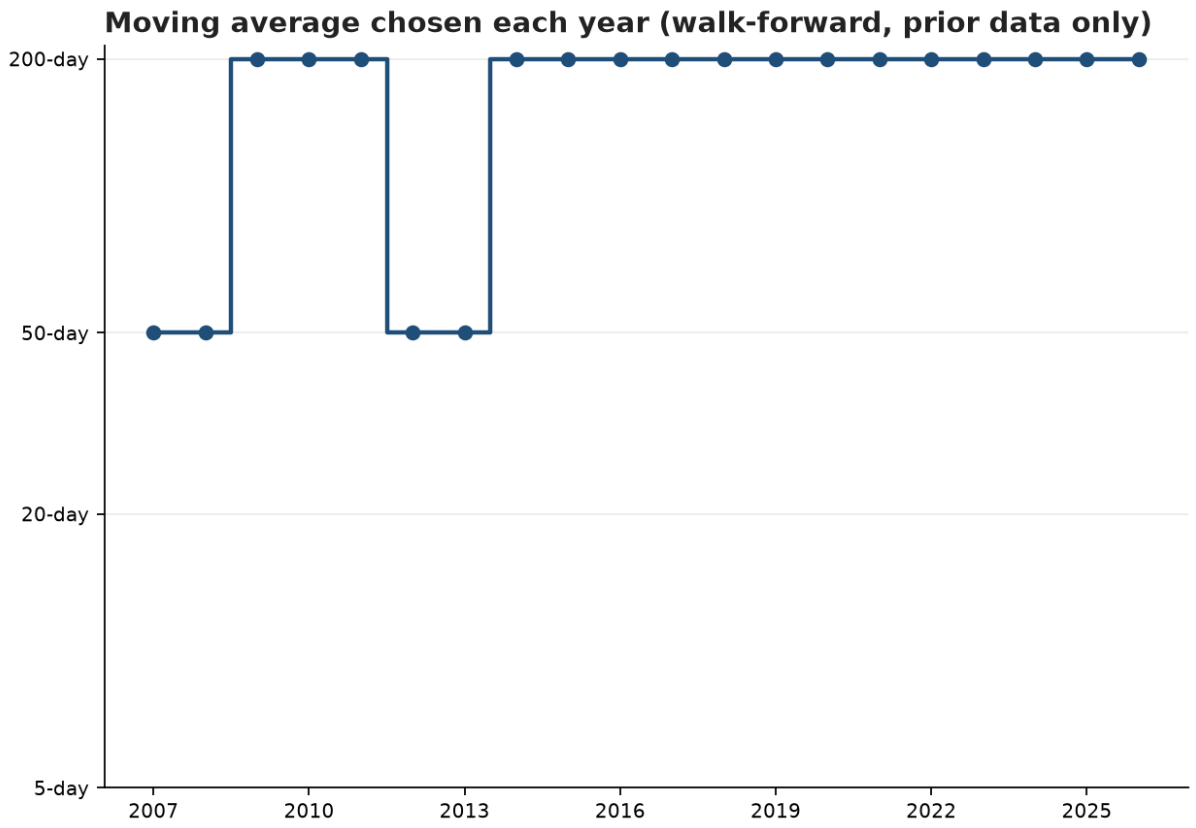
- Verdict since 2011: SPY led on return and Sharpe (14.1% CAGR, 0.77). Neither the MA strategy nor any momentum lookback beat it on either measure; momentum figures exclude trading costs.
- MA strategy since 2011: 9.3% CAGR net of costs, Sharpe 0.67, max drawdown -26.4% vs SPY -33.7%. Highest Sharpe of the systematic approaches; tied with 3-Month momentum (-25.9%) for shallowest drawdown.
- Momentum since 2011: CAGR 7.6% to 11.2%, Sharpe 0.40 to 0.59, max drawdown -25.9% to -43.8%. 3-Month was the strongest lookback (11.2% CAGR before costs, 10.9% if charged MA costs).
- MA strategy since 2007: max drawdown -26.4% vs SPY -55.2%, but CAGR 7.6% vs 10.9% and Sharpe 0.53 vs 0.55. A 60/40 portfolio beat it on both CAGR (8.4%) and Sharpe (0.63).
- Recent windows: the MA strategy trailed SPY on Sharpe over 1Y (0.78 vs 1.0) and 5Y (0.39 vs 0.57), and had a deeper 5Y drawdown (-26.4% vs -24.5%).
- The case: a tactical SPY/Treasury allocation that has historically cut deep equity drawdowns at a real cost in upside; a risk-managed complement, not an SPY replacement.

The Problem: Deep Equity Drawdowns

- Since 2007, SPY's maximum drawdown was -55.2%, and its worst rolling 1-year return was -47.4%.
- In 2008 SPY returned -36.8%; a 60/40 portfolio returned -16.7%.
- In 2022 stocks and bonds fell together: SPY -18.2%, 60/40 -16.4%, so traditional diversification offered limited relief.
- Large losses are hard to recover from and can force investors to sell at the wrong time.
- The MA strategy aims to reduce exposure to prolonged equity declines by rotating into intermediate Treasuries (IEF).

How the MA Strategy Works: Walk-Forward Rule Selection

- Daily rule: hold SPY when SPY is above its moving average; otherwise hold IEF (intermediate Treasuries).
- Each January the average length (5, 20, 50 or 200 days) is re-selected using only data available at that time.
- Over 20 years of walk-forward selection, the 200-day average was chosen 16 times and the 50-day 4 times.
- Results are net of modeled trading costs. Since 2007: 158 switches, 77.0% of time in SPY.
- Current allocation: 100.0% SPY.



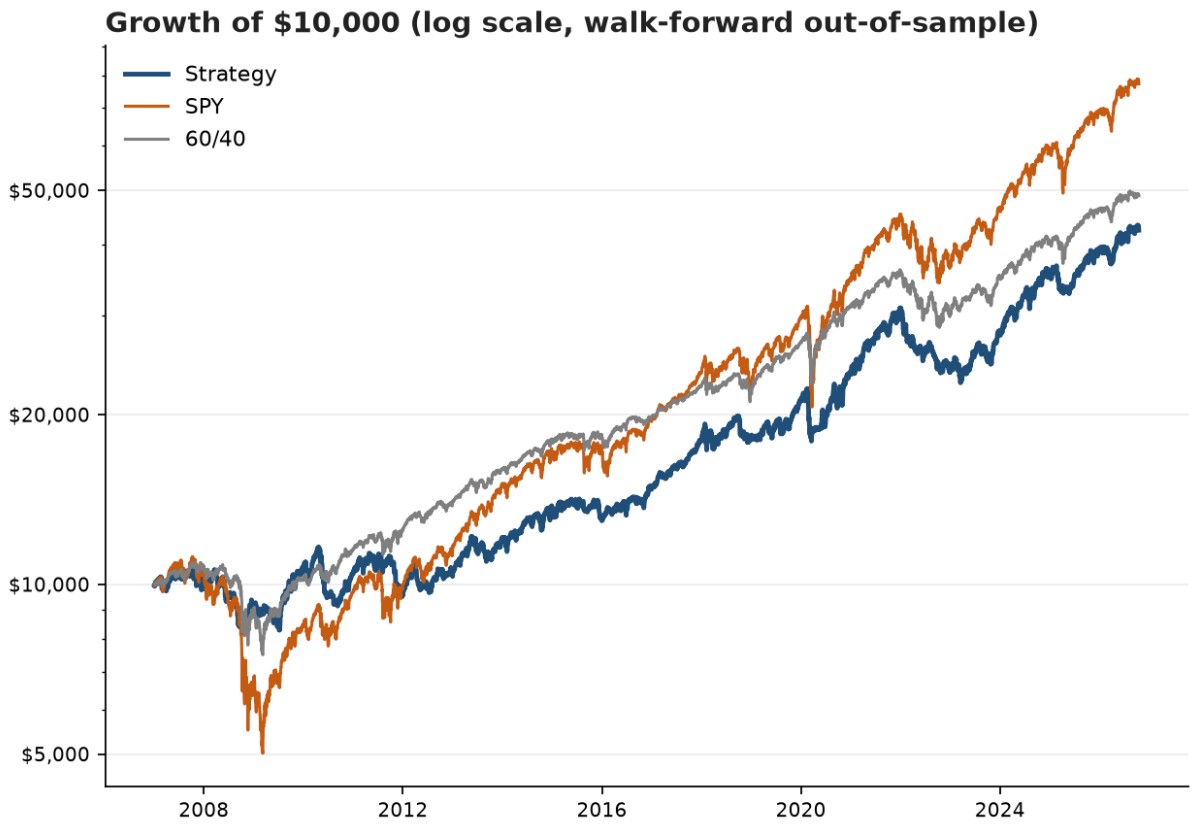
Context: Each Moving Average Over the Full Period (IN-SAMPLE)

Rule (IN-SAMPLE)	CAGR	Volatility	Sharpe	Max DD	Switches	% in SPY	Years chosen
5-day MA switch	-3.5%	12.9%	-0.33	-66.5%	1330	59.4%	0
20-day MA switch	4.0%	12.3%	0.26	-37.1%	570	66.3%	0
50-day MA switch	6.0%	12.0%	0.42	-29.8%	348	70.2%	4
200-day MA switch	9.4%	12.5%	0.66	-26.4%	112	79.1%	16

IN-SAMPLE (2007-01-03 to 2026-10-16): these rows use hindsight. Picking the best row after the fact would overstate results. The strategy instead picks a moving average each January using only earlier data, and that walk-forward result is what the performance pages show. "Years chosen" counts how many years the walk-forward selected each rule.

MA Strategy vs SPY Since 2007: Growth of $10,000

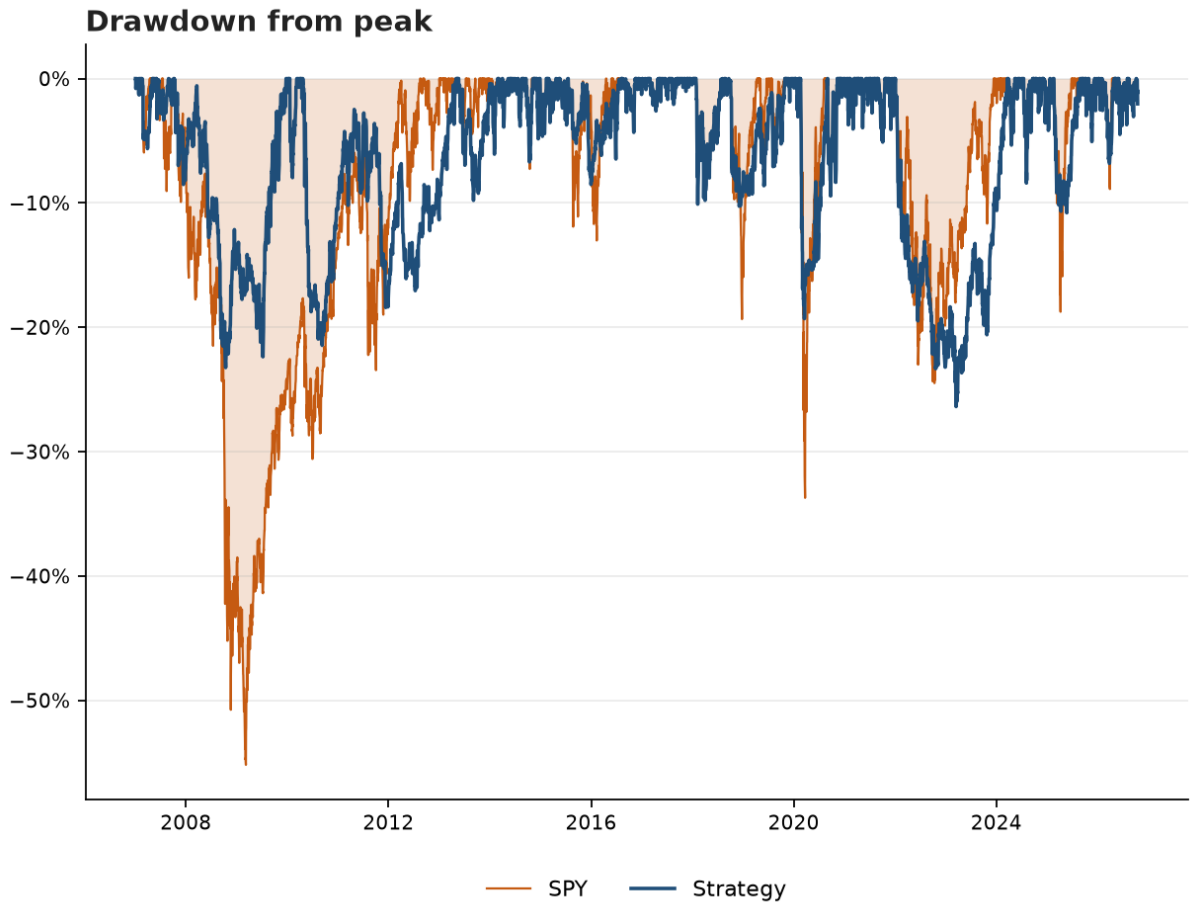
- $10,000 grew to $42,495 in the MA strategy vs $77,253 in SPY and $48,905 in 60/40.
- CAGR 7.6% vs SPY 10.9%, with volatility of 12.4% vs 19.5%.
- Sharpe 0.53 vs SPY 0.55, broadly similar; 60/40 was higher at 0.63.
- Up capture 54.9% and down capture 53.5%: the strategy captured only part of SPY's gains as well as its losses.
- Beta 0.29 and correlation 0.45 to SPY over the full period.



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Risk and Drawdown Protection vs SPY

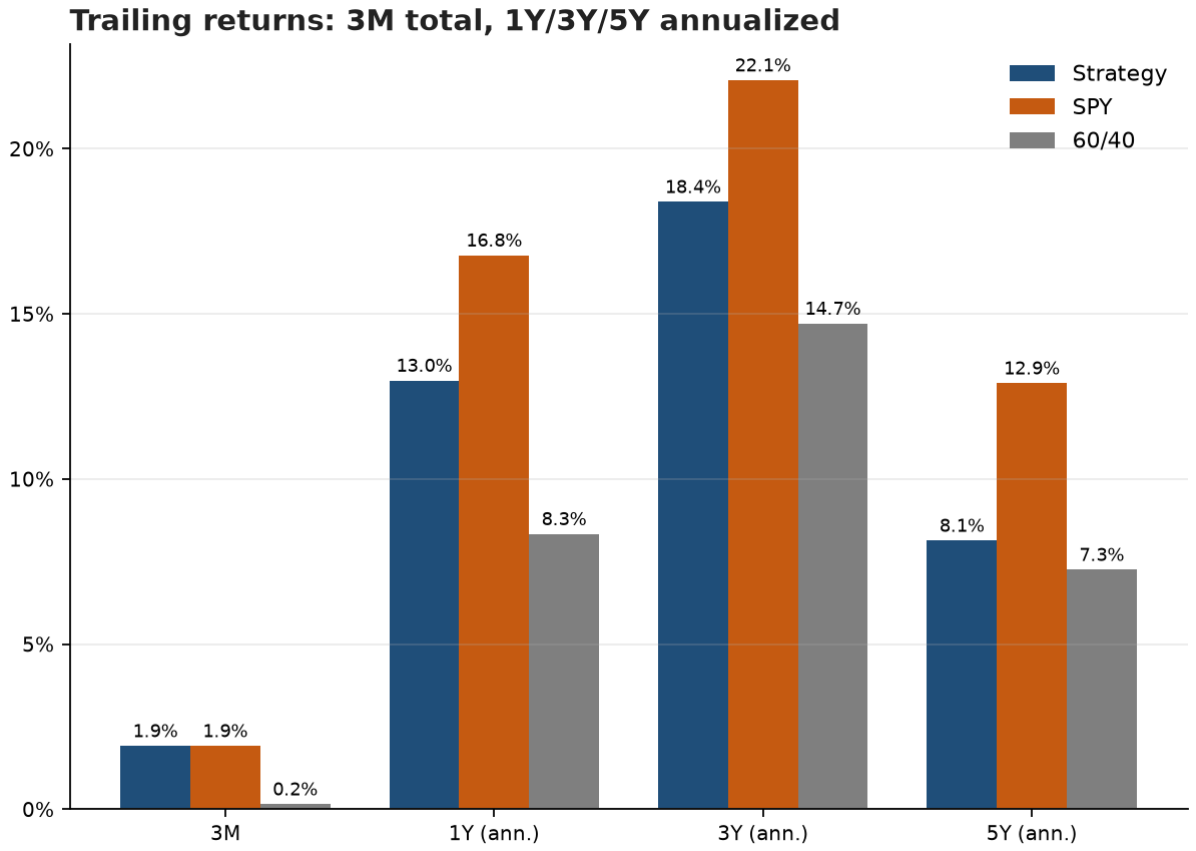
- Since 2007: max drawdown -26.4% vs SPY -55.2%; calmar 0.29 vs 0.20.
- Worst rolling 1-year return -23.1% vs SPY -47.4%.
- 2008: strategy -6.8% vs SPY -36.8%, as the rule moved into Treasuries.
- Protection was not consistent: in 2022 the strategy lost -22.7% vs SPY -18.2%, as IEF also fell.
- Over 5Y the drawdown was deeper than SPY's (-26.4% vs -24.5%).



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MA Strategy Performance by Period: 3M, 1Y, 3Y, 5Y

- 3M: 1.9% vs SPY 1.9%, identical, as the strategy held SPY 100.0% of the time with no switches.
- 1Y: 13.0% vs SPY 16.8%; Sharpe 0.78 vs 1.0; max drawdown -6.8% vs -8.9%.
- 3Y (annualized): 18.4% vs SPY 22.1%; Sharpe 1.11 vs 1.1 (a tie); max drawdown -10.8% vs -18.8%.
- 5Y (annualized): 8.1% vs SPY 12.9%; Sharpe 0.39 vs 0.57; max drawdown -26.4% vs -24.5%.
- 60/40 for reference: 1Y 8.3%, 3Y 14.7%, 5Y 7.3% annualized.



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Risk and Return by Period: Strategy vs SPY

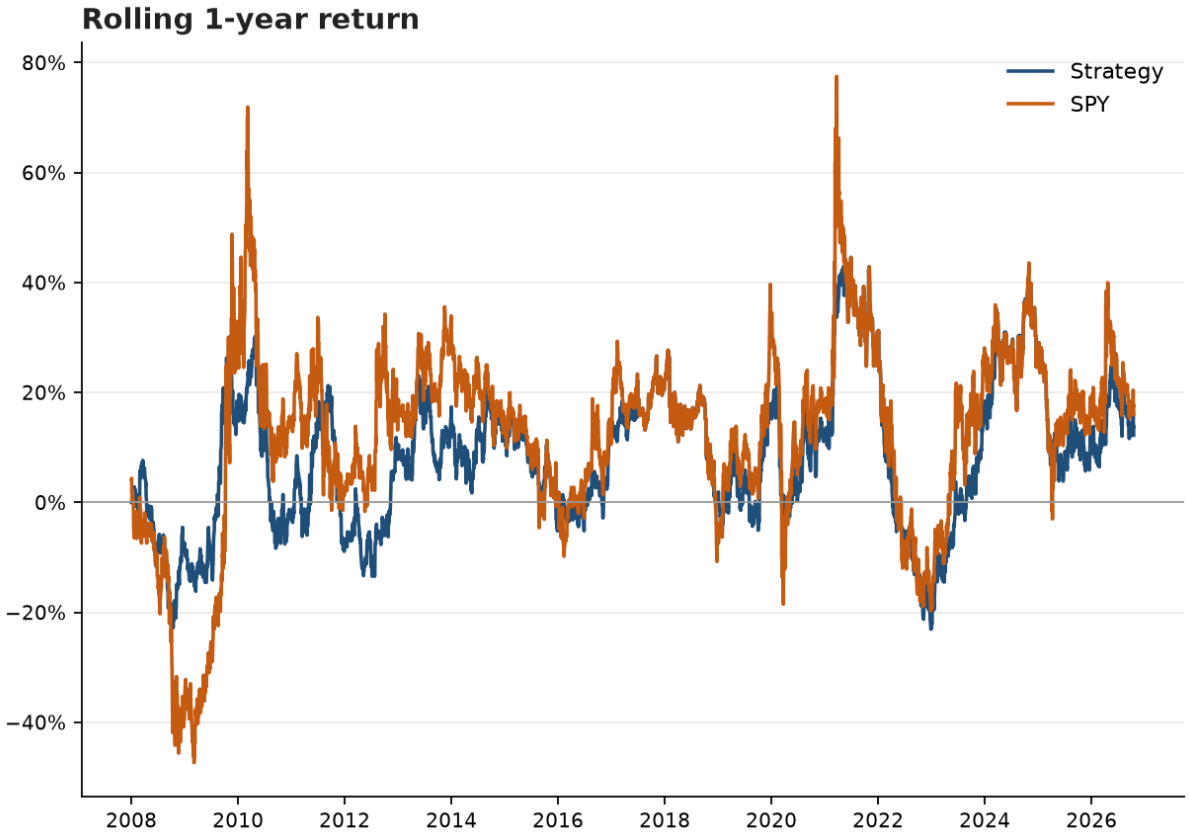
Metric	3M Strat.	3M SPY	1Y Strat.	1Y SPY	3Y Strat.	3Y SPY	5Y Strat.	5Y SPY	SI Strat.	SI SPY
Return (3M total, else annualized)	**1.9%**	**1.9%**	**13.0%**	**16.8%**	**18.4%**	**22.1%**	**8.1%**	**12.9%**	**7.6%**	**10.9%**
Total return (cumulative)	1.9%	1.9%	13.0%	16.8%	65.6%	81.5%	47.6%	83.0%	324.9%	672.5%
Volatility (ann.)	11.1%	11.1%	11.9%	12.7%	12.0%	15.3%	12.3%	17.1%	12.4%	19.5%
Sharpe ratio	n/a	n/a	0.78	1.00	1.11	1.10	0.39	0.57	0.53	0.55
Sortino ratio	n/a	n/a	1.11	1.47	1.57	1.63	0.54	0.83	0.72	0.77
Max drawdown	-3.1%	-3.1%	-6.8%	-8.9%	-10.8%	-18.8%	-26.4%	-24.5%	-26.4%	-55.2%
Calmar ratio	n/a	n/a	1.90	1.89	1.70	1.18	0.31	0.53	0.29	0.20
Beta vs SPY	1.00	1.00	0.87	1.00	0.58	1.00	0.46	1.00	0.29	1.00
Correlation vs SPY	1.00	1.00	0.94	1.00	0.74	1.00	0.65	1.00	0.45	1.00
Up capture	100.0%	100.0%	91.7%	100.0%	83.3%	100.0%	65.7%	100.0%	54.9%	100.0%
Down capture	100.0%	100.0%	94.6%	100.0%	83.3%	100.0%	65.9%	100.0%	53.5%	100.0%
% days positive	44.6%	44.6%	52.6%	52.6%	56.2%	56.4%	53.6%	53.7%	54.1%	55.0%
Best day	1.8%	1.8%	1.9%	2.9%	2.5%	10.5%	2.5%	10.5%	4.4%	14.5%
Worst day	-1.5%	-1.5%	-2.6%	-2.6%	-3.0%	-5.9%	-3.0%	-5.9%	-5.8%	-10.9%
Switches / % time in SPY	0 / 100.0%		2 / 95.2%		8 / 92.2%		30 / 78.2%		158 / 77.0%	

60/40 benchmark	3M	1Y	3Y	5Y	SI
Return (3M total, else ann.)	0.2%	8.3%	14.7%	7.3%	8.4%
Total return (cumulative)	0.2%	8.3%	50.7%	41.7%	389.0%
Volatility (ann.)	7.8%	8.5%	9.7%	11.0%	11.2%
Sharpe ratio	n/a	0.55	1.02	0.35	0.63
Max drawdown	-2.8%	-6.0%	-10.6%	-21.0%	-31.4%
Calmar ratio	n/a	1.39	1.39	0.35	0.27

Periods end on the last completed session. 3M: 2026-07-17 to 2026-10-16, 1Y: 2025-10-17 to 2026-10-16, 3Y: 2023-10-17 to 2026-10-16, 5Y: 2021-10-18 to 2026-10-16, SI: 2007-01-03 to 2026-10-16. SI = since the walk-forward start. Ratios use the 13-week T-bill as the risk-free rate. Current allocation: 100.0% SPY.

Rolling 1-Year Returns: MA Strategy vs SPY

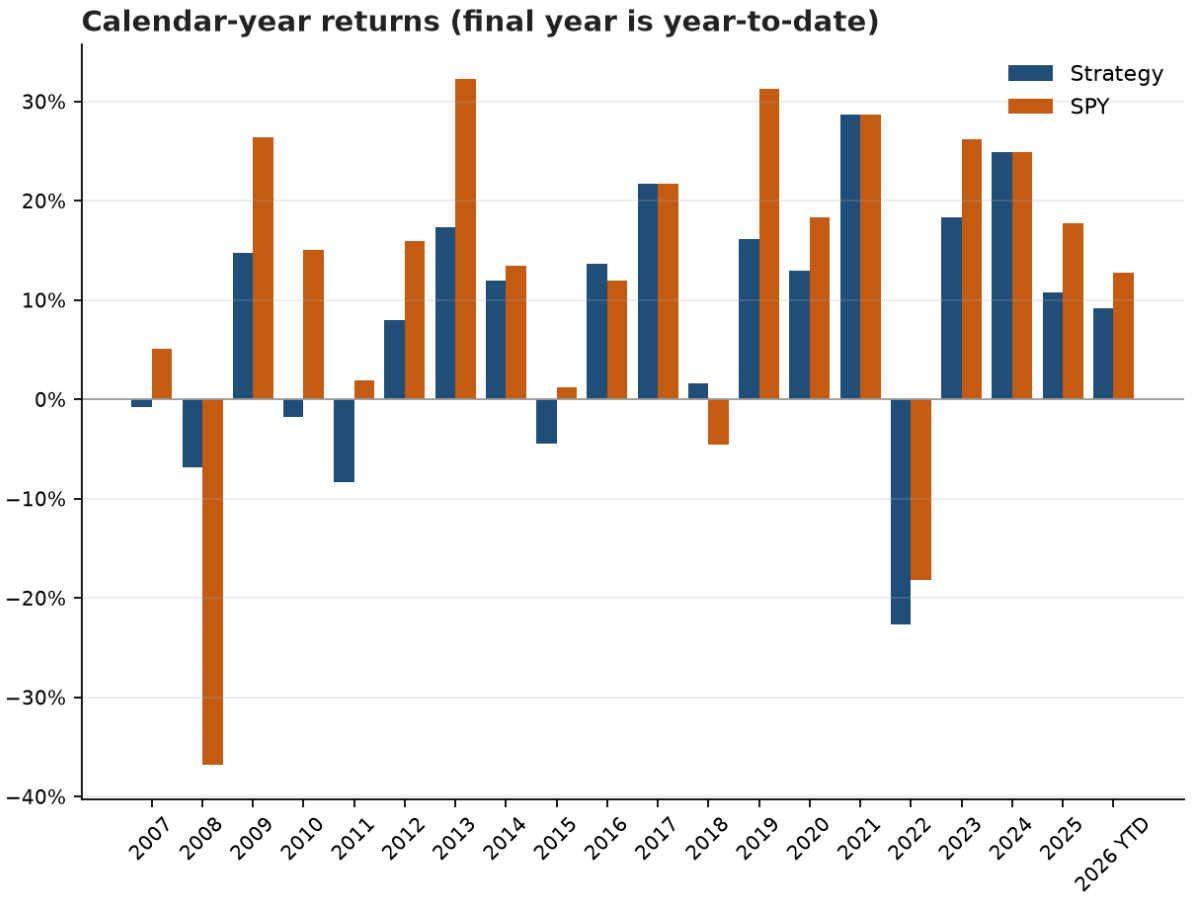
- Across 4,728 rolling 1-year periods, the strategy beat SPY in 18.7% of them; median 1-year excess return was -4.1%.
- Best rolling year 44.6% vs SPY 77.5%; worst rolling year -23.1% vs SPY -47.4%.
- The profile is asymmetric: it has lagged in most periods and added value mainly in severe, prolonged declines.



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When the MA Strategy Lags: Honest Trade-Offs

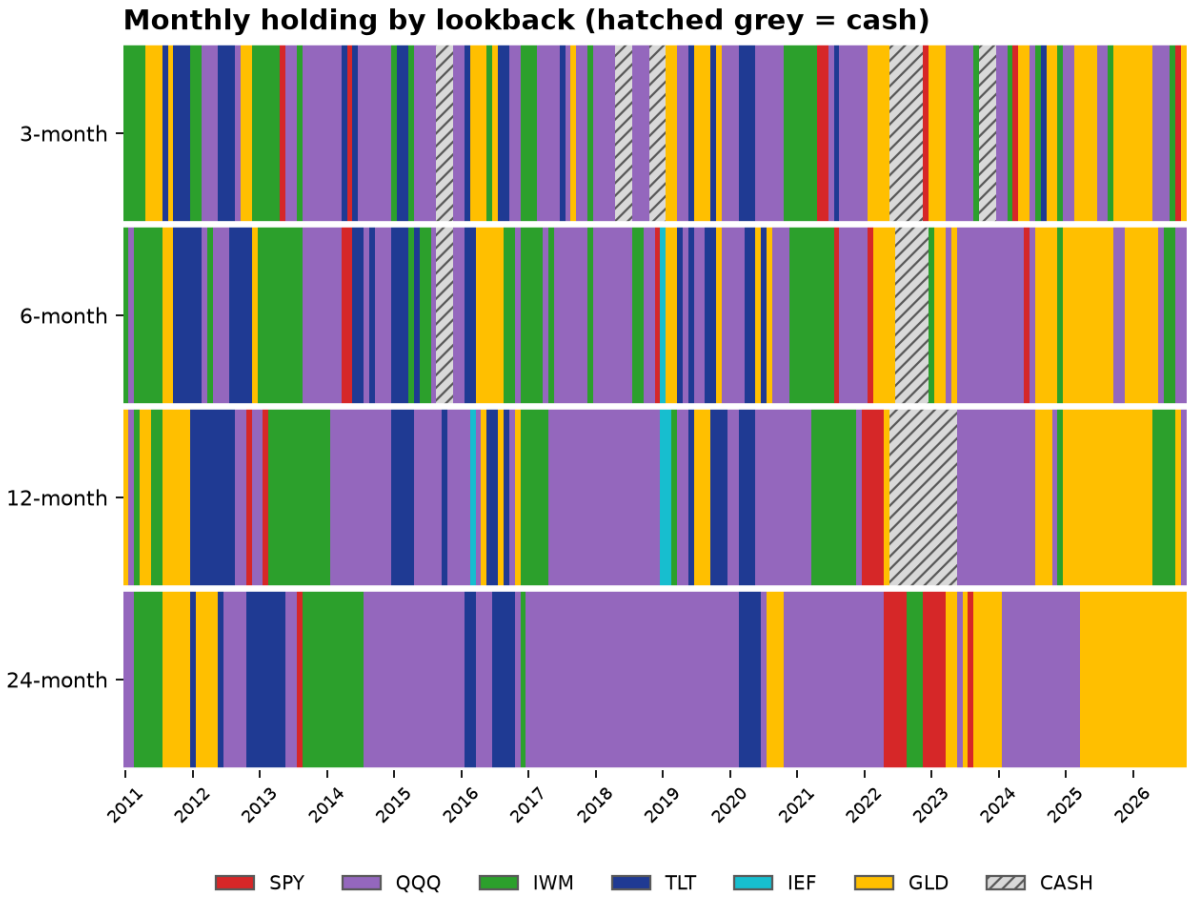
- 1Y: worse than SPY on Sharpe (0.78 vs 1.0).
- 5Y: worse than SPY on max drawdown (-26.4% vs -24.5%) and Sharpe (0.39 vs 0.57).
- Since 2007: worse than SPY on Sharpe (0.53 vs 0.55), and worse than 60/40 on CAGR (7.6% vs 8.4%) and Sharpe (0.53 vs 0.63).
- Whipsaw years hurt: 2011 -8.3% vs SPY 1.9%; 2015 -4.5% vs SPY 1.2%; 2010 -1.8% vs SPY 15.1%.
- When bonds and stocks fall together the IEF leg offers little shelter, as in 2022 (-22.7% vs SPY -18.2%).
- 2026 YTD (through 2026-10-16): 9.1% vs SPY 12.8%.



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Multi-Asset Momentum: Design and Holdings Since 2011

- Four variants using 3, 6, 12 and 24-month lookbacks rank SPY, QQQ, TLT, GLD, IWM and IEF by trailing return each month-end.
- Each holds 100% of the top-ranked asset for the next month; if all six are negative it holds cash for 3 months.
- QQQ was the most-held asset in every variant, from 36.8% (6-Month) to 53.7% (24-Month) of months.
- Results run from January 2011 and exclude trading costs; the MA strategy is net of modeled costs.
- Current holdings: 3-Month and 24-Month in GLD; 6-Month and 12-Month in QQQ.



3-Month Momentum: Pros and Cons

- Pro: Shallower max drawdown than SPY since 2011 (-25.9% vs -33.7%), the shallowest of the four lookbacks.
- Pro: Higher CAGR than the MA strategy even if charged MA costs (10.9% vs 9.3%); 11.2% before costs.
- Pro: Low correlation to SPY (0.28), with standout years such as 2020 (52.1% vs SPY 18.3%).
- Con: Worse than SPY since 2011 on CAGR (11.2% before costs vs 14.1%).
- Con: Worse than SPY since 2011 on Sharpe (0.59 vs 0.77), with higher volatility (18.0% vs 17.0%).
- Con: Most active variant (80 switches, 18 cash months); weak years include 2015 (-14.5% vs SPY 1.2%).

6-Month Momentum: Pros and Cons

- Pro: Strong recent run before costs: 3Y 25.1% vs SPY 22.1% and 5Y 15.9% vs SPY 12.9%, annualized.
- Pro: Large gains in trending years, e.g. 2025 54.7% vs SPY 17.7%.
- Con: Worse than SPY since 2011 on CAGR (7.6% before costs vs 14.1%); the lowest of the four lookbacks.
- Con: Worse than SPY since 2011 on Sharpe (0.40 vs 0.77).
- Con: Worse than SPY since 2011 on max drawdown (-43.8% vs -33.7%), the deepest of any approach shown.
- Con: Below the MA strategy on CAGR even if charged MA costs (7.4% vs 9.3%).

12-Month Momentum: Pros and Cons

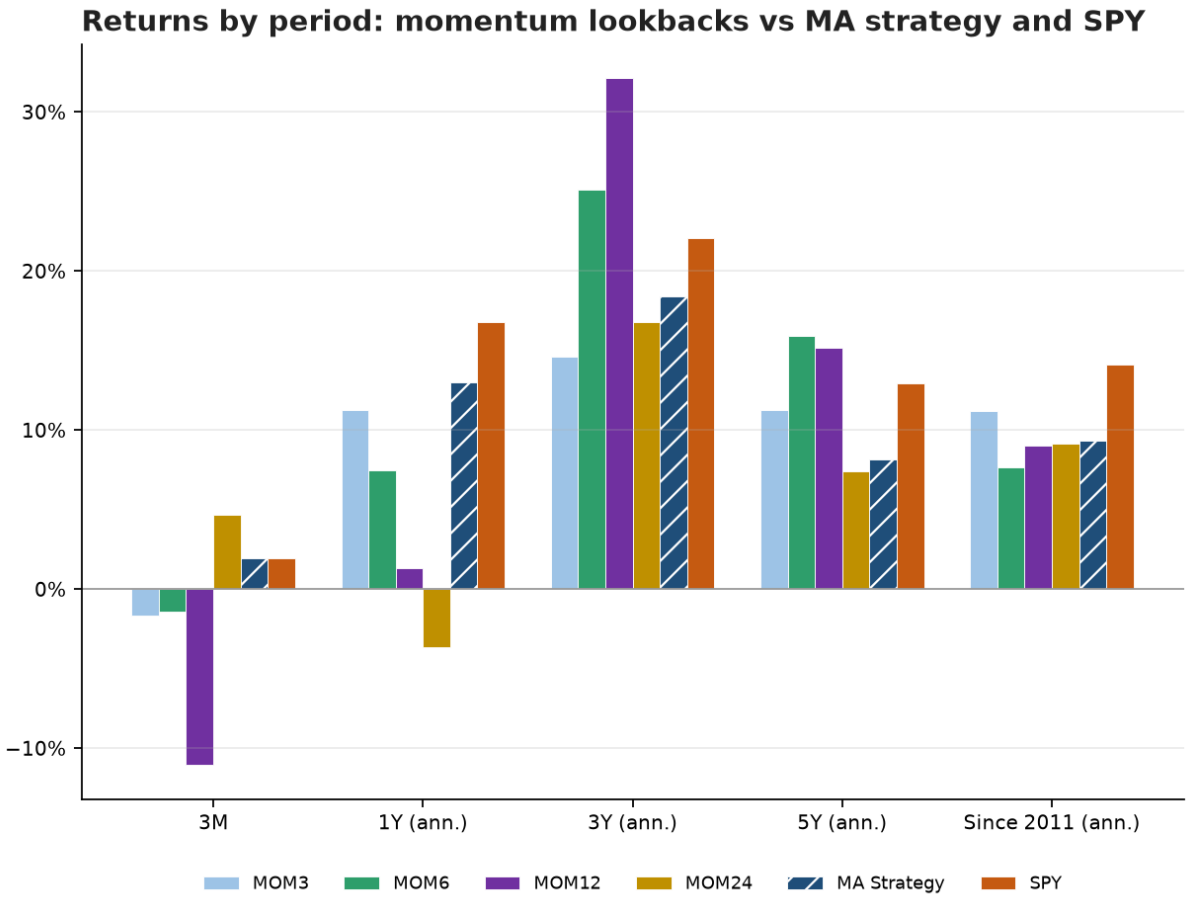
- Pro: Strongest 3Y return before costs: 32.1% vs SPY 22.1% annualized; 5Y 15.1% vs 12.9%.
- Pro: Exceptional 2025 (63.7% vs SPY 17.7%) and 2020 (38.1% vs 18.3%).
- Con: Worse than SPY since 2011 on CAGR (9.0% before costs vs 14.1%).
- Con: Worse than SPY since 2011 on Sharpe (0.48 vs 0.77); max drawdown was similar (-33.3% vs -33.7%).
- Con: Sharp recent reversal: 3M -11.0% vs SPY 1.9%; 1Y 1.3% vs 16.8%. Bad years include 2016 (-19.6% vs 12.0%).

24-Month Momentum: Pros and Cons

- Pro: Lowest turnover (31 switches since 2011), so modeled costs matter least: 9.1% CAGR before costs, 9.0% if charged MA costs.
- Pro: Strong in some years SPY led, e.g. 2019 (39.0% vs 31.2%) and 2017 (32.7% vs 21.7%).
- Con: Worse than SPY since 2011 on CAGR (9.1% before costs vs 14.1%).
- Con: Worse than SPY since 2011 on Sharpe (0.46 vs 0.77).
- Con: Worse than SPY since 2011 on max drawdown (-34.2% vs -33.7%), with no use of cash (0 cash months).
- Con: Highest SPY sensitivity of the lookbacks (up capture 78.2%, down capture 79.1%); 2022 -26.5% vs SPY -18.2%.

Momentum vs MA Strategy vs SPY Since 2011

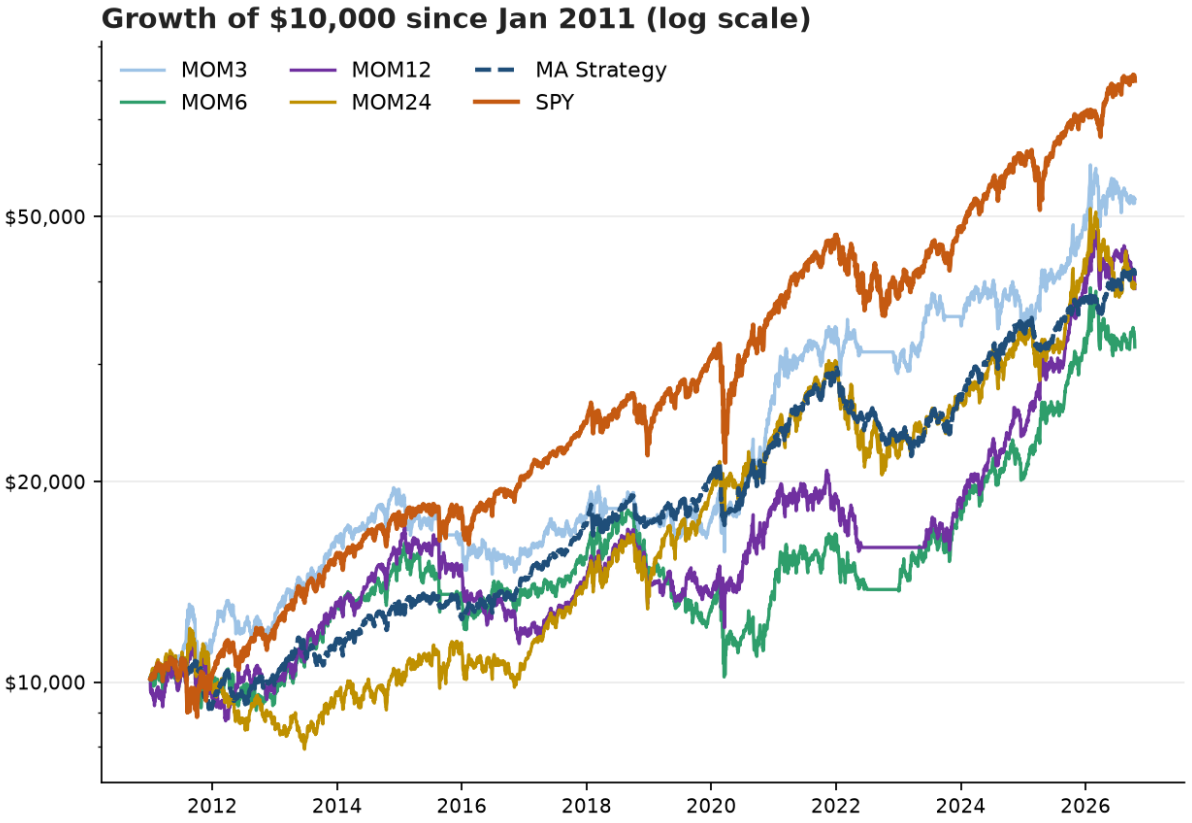
- CAGR: SPY 14.1%; MA strategy 9.3% net of costs; momentum 7.4% to 10.9% if charged MA costs (7.6% to 11.2% before costs, as reported).
- Sharpe: SPY 0.77 led; MA strategy 0.67; momentum 0.40 to 0.59.
- Max drawdown: 3-Month momentum -25.9% and MA strategy -26.4% (a tie) vs SPY -33.7%; other lookbacks -33.3% to -43.8%.
- Volatility: MA strategy 12.1% vs SPY 17.0% and momentum 18.0% to 19.6%.
- Momentum excludes trading costs; the MA strategy is net of modeled costs. Charging MA costs lowers momentum CAGR by up to 0.3 points.
- Growth of $10,000: SPY $79,961; 3-Month momentum $53,177; MA strategy $40,753.



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Momentum Growth vs SPY Since 2011

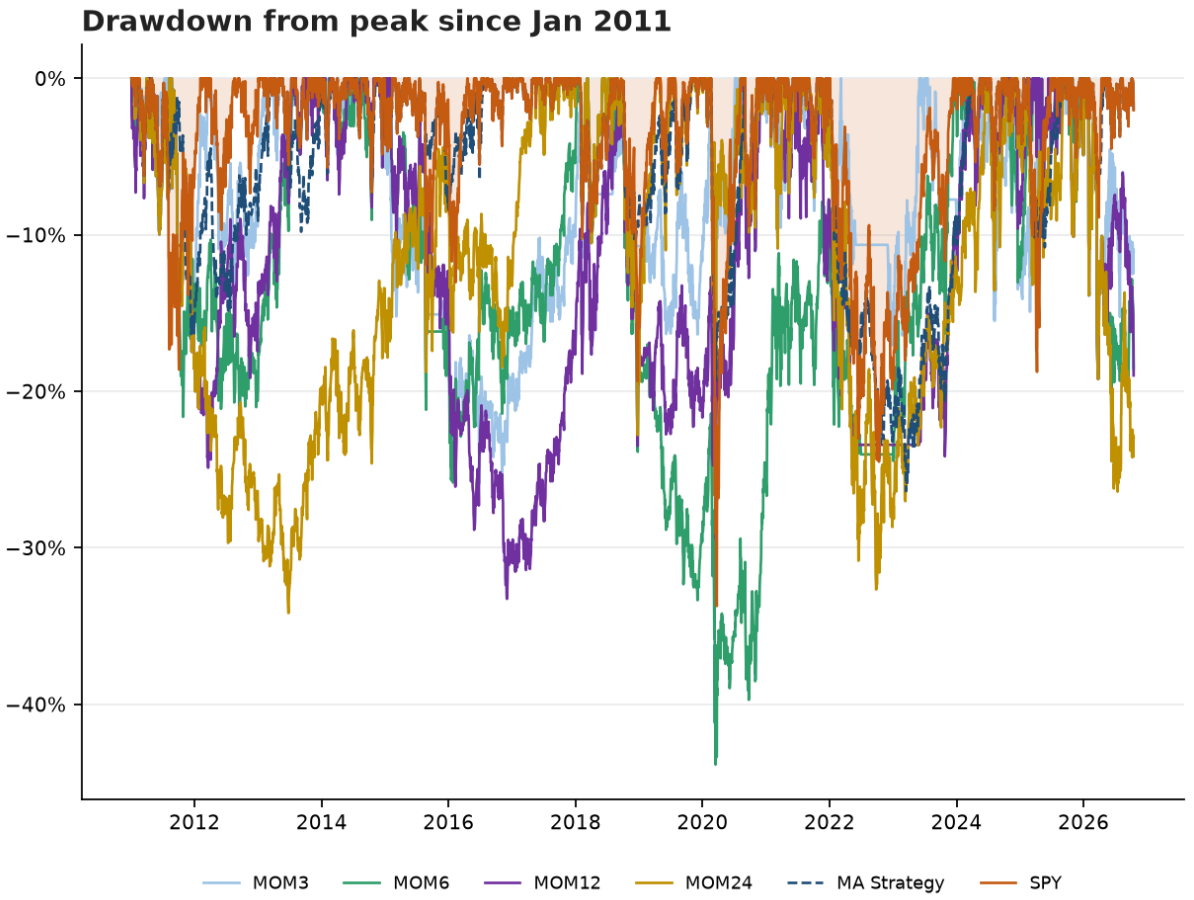
- Growth of $10,000 before costs: 3-Month $53,177, 24-Month $39,609, 12-Month $38,981, 6-Month $31,853, vs SPY $79,961.
- Returns arrive in bursts: 2025 ranged from 26.3% to 63.7% across lookbacks vs SPY 17.7%.
- Long flat or losing stretches also occurred, e.g. 2016 from -19.6% to 1.3% vs SPY 12.0%.
- Lookback choice mattered a lot: since-2011 CAGR ranged from 7.6% to 11.2% before costs.



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Momentum Drawdowns vs SPY Since 2011

- Max drawdown: 3-Month -25.9%, 12-Month -33.3%, 24-Month -34.2%, 6-Month -43.8%, vs SPY -33.7% and MA strategy -26.4%.
- Single-asset concentration drives the risk: each variant holds 100% of one ETF, so volatility ran 18.0% to 19.6% vs SPY 17.0%.
- 2022 was hard for every approach: momentum -12.4% to -26.5% vs SPY -18.2% and MA strategy -22.7%.



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Head to Head Since 2011: Momentum vs MA Strategy vs SPY

Return (3M total, else annualized)	3M	1Y	3Y	5Y	Since 2011
3-month momentum	-1.7%	11.2%	14.6%	11.3%	11.2%
6-month momentum	-1.4%	7.5%	25.1%	15.9%	7.6%
12-month momentum	-11.0%	1.3%	32.1%	15.1%	9.0%
24-month momentum	4.7%	-3.6%	16.8%	7.4%	9.1%
MA trend strategy	1.9%	13.0%	18.4%	8.1%	9.3%
SPY (buy and hold)	1.9%	16.8%	22.1%	12.9%	14.1%

Since 2011	CAGR	Total	Vol.	Sharpe	Sortino	Max DD	Calmar	Beta	Corr.	Up cap.	Down cap.
3-month momentum	11.2%	431.8%	18.0%	0.59	0.83	-25.9%	0.43	0.30	0.28	55.7%	50.8%
6-month momentum	7.6%	218.5%	19.0%	0.40	0.55	-43.8%	0.17	0.53	0.47	60.4%	59.9%
12-month momentum	9.0%	289.8%	18.4%	0.48	0.66	-33.3%	0.27	0.40	0.37	62.6%	61.1%
24-month momentum	9.1%	296.1%	19.6%	0.46	0.64	-34.2%	0.27	0.55	0.48	78.2%	79.1%
MA trend strategy	9.3%	307.5%	12.1%	0.67	0.91	-26.4%	0.35	0.41	0.58	65.3%	65.1%
SPY (buy and hold)	14.1%	699.6%	17.0%	0.77	1.08	-33.7%	0.42	1.00	1.00	100.0%	100.0%

Common period 2011-01-03 to 2026-10-16. Momentum: hold the top trailing performer of SPY, QQQ, TLT, GLD, IWM and IEF each month, or cash (0%) for 3 months when all are negative. **Momentum excludes trading costs; the MA strategy is net of modeled costs.** Ratios use the 13-week T-bill as the risk-free rate. Beta, correlation and capture are measured vs SPY.

Momentum Detail: Calendar Years, Holdings and Activity

Year	3M	6M	12M	24M	MA	SPY
2011	13.8%	1.4%	-4.1%	-0.8%	-8.3%	1.9%
2012	7.9%	-7.8%	2.4%	-12.1%	8.0%	16.0%
2013	35.0%	40.6%	34.4%	10.9%	17.4%	32.3%
2014	15.6%	11.9%	18.1%	6.1%	12.0%	13.5%
2015	-14.5%	-9.4%	-5.7%	9.4%	-4.5%	1.2%
2016	-5.0%	1.3%	-19.6%	-6.9%	13.6%	12.0%
2017	13.2%	13.2%	19.3%	32.7%	21.7%	21.7%
2018	-0.2%	-3.7%	-0.1%	-0.1%	1.6%	-4.6%
2019	-0.9%	-13.2%	-4.8%	39.0%	16.2%	31.2%
2020	52.1%	5.4%	38.1%	22.3%	12.9%	18.3%
2021	27.3%	22.3%	2.2%	27.4%	28.7%	28.7%
2022	-12.4%	-16.4%	-15.8%	-26.5%	-22.7%	-18.2%
2023	19.9%	31.6%	18.3%	21.5%	18.3%	26.2%
2024	-1.4%	11.4%	24.6%	21.6%	24.9%	24.9%
2025	36.9%	54.7%	63.7%	26.3%	10.8%	17.7%
2026 YTD	11.3%	1.9%	1.4%	-3.6%	9.1%	12.8%

Since 2011	Months	Switches	Cash mo.	Holding now
3-month	190	80	18	GLD
6-month	190	71	9	QQQ
12-month	190	48	12	QQQ
24-month	190	31	0	GLD

% of months held	SPY	QQQ	TLT	GLD	IWM	IEF	CASH
3-month	3.7%	37.9%	11.6%	21.1%	16.3%	-	9.5%
6-month	3.2%	36.8%	13.2%	20.5%	21.1%	0.5%	4.7%
12-month	3.2%	41.1%	12.1%	18.4%	17.4%	1.6%	6.3%
24-month	5.3%	53.7%	10.0%	20.5%	10.5%	-	-

If charged 5 bps per trade	Trades*	CAGR as shown	CAGR with costs
3-month	81	11.2%	10.9%
6-month	72	7.6%	7.4%
12-month	49	9.0%	8.8%
24-month	32	9.1%	9.0%

Growth of $10,000 since 2011	Value
3-month momentum	$53,177
6-month momentum	$31,853
12-month momentum	$38,981
24-month momentum	$39,609
MA trend strategy	$40,753
SPY (buy and hold)	$79,961

Calendar-year returns are total returns; the final year is year-to-date through 2026-10-16. 3M/6M/12M/24M = momentum lookbacks; MA = moving-average trend strategy. Momentum excludes trading costs. *Trades = switches plus the initial purchase in January 2011.

Portfolio Fit: Who the MA Strategy Suits

- Suits investors who prioritize limiting severe equity drawdowns and accept lower long-run returns than SPY (since 2007 CAGR 7.6% vs 10.9%).
- Can serve as a rules-based sleeve alongside core equity: beta 0.29 and correlation 0.45 to SPY since 2007.
- Not a substitute for a 60/40 allocation on returns alone: 60/40 had higher CAGR (8.4%) and Sharpe (0.63) since 2007.
- Momentum variants are more concentrated and volatile (18.0% to 19.6% vs SPY 17.0% since 2011) and suit only a small, risk-tolerant satellite allocation.
- Expect long stretches of lagging SPY: the strategy beat SPY in only 18.7% of rolling 1-year periods.

Hypothetical backtested performance. Not actual trading; not net of fees. MA strategy results are net of modeled trading costs; momentum results exclude trading costs. Past performance does not guarantee future results. See Important Disclosures.

Next Steps

- Request the full methodology document, including the walk-forward selection procedure and trading-cost model.
- Request access to the underlying daily return series to run your own portfolio-level analysis.
- Discuss sizing and implementation within your existing allocation with your advisor or investment committee.

For discussion purposes only. Not an offer or solicitation.

Meet the Digital Investment Team (AI-generated, fictional)

Margaret Okonkwo-Hale

Lead Portfolio Manager

Focus: Portfolio construction and strategy oversight

Certifications: CFA, CAIA | **Experience:** 29 yrs

Education

MBA, Finance, University of Chicago Booth School of Business (2001)
BA, Economics, Williams College (1995)

Experience

Senior Portfolio Manager, Tactical Allocation, Harborview Capital Partners (2012-2024)
Portfolio Manager, Ashgrove Asset Advisors (2004-2012)
Research Associate, Lindenfield Securities (1997-2004)

Margaret Okonkwo-Hale is an AI-generated fictional persona who leads the team as Portfolio Manager. Her invented career spans nearly three decades in tactical asset allocation at fictional firms, moving from equity research into multi-asset portfolio management. On this team she owns final decisions on strategy design, allocation rules, and how the trend rotation and momentum strategies fit together in client portfolios. Her distinctive strength is disciplined rule adherence: she insists that every systematic signal be documented, stress-tested, and followed without discretionary overrides during turbulent markets.

AI-generated fictional persona

Daniel Reyes-Whitford

Senior Quantitative Research Analyst

Focus: Quantitative research and signal design

Certifications: CQF | **Experience:** 15 yrs

Education

PhD, Applied Mathematics, Cornell University (2011)
BS, Physics, University of Texas at Austin (2005)

Experience

Quantitative Researcher, Quillstone Quantitative (2015-2025)
Research Scientist, Meridian Bluff Analytics (2011-2015)

Daniel Reyes-Whitford is an AI-generated fictional persona serving as the team's senior quantitative researcher. Trained as an applied mathematician, his invented background includes a decade building systematic signals at fictional quant boutiques. He owns the design and ongoing review of the moving-average rules behind the SPY/IEF rotation and the lookback and ranking logic in the multi-asset momentum model. His distinctive strength is skepticism toward overfitting; he favors simple, robust parameter choices and routinely tests how sensitive each signal is to small changes in its settings.

AI-generated fictional persona

Priya Natarajan

Risk Management Analyst

Focus: Risk management, drawdown and exposure monitoring

Certifications: FRM, PRM | **Experience:** 12 yrs

Education

MS, Financial Engineering, Columbia University (2014)
BTech, Electrical Engineering, Indian Institute of Technology Madras (2012)

Experience

Risk Analyst, Corvane Risk Partners (2018-2025)
Associate, Market Risk, Stillwater Ridge Investments (2014-2018)

Priya Natarajan is an AI-generated fictional persona who leads risk management for the team. An engineer turned financial engineer, her invented career includes market risk roles at fictional investment firms. She owns drawdown monitoring, volatility and correlation analysis across SPY, QQQ, TLT, GLD, IWM and IEF, and the scenario tests that examine how both strategies behave during sharp reversals. Her distinctive strength is translating abstract risk statistics into clear, practical limits and plain-language explanations that the portfolio manager and investors can act on.

AI-generated fictional persona

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Digital Investment Team, continued (AI-generated, fictional)

Thomas Achebe-Lindqvist

Macro and Rates Analyst
Focus: Macro and interest rates, Treasury exposure (IEF/TLT)

Certifications: CFA | **Experience:** 17 yrs

Education
MA, Economics, London School of Economics (2009)
BA, History and Economics, University of Michigan (2007)

Experience
Rates Strategist, Northgate Fixed Income Advisors (2014-2025)
Economist, Pellham Economic Research (2009-2014)

Thomas Achebe-Lindqvist is an AI-generated fictional persona covering macro conditions and interest rates. With an economics background and an invented career as a rates strategist at fictional firms, he focuses on the Treasury side of both strategies, especially IEF and TLT. He owns analysis of duration risk, yield-curve regimes, and periods when stocks and bonds fall together, which can challenge defensive rotation. His distinctive strength is historical perspective: he places current rate environments in the context of past inflation and policy cycles.

AI-generated fictional persona

Kenji Morales

Data Engineering and Backtest Analyst
Focus: Market data engineering and backtest integrity

Certifications: None | **Experience:** 8 yrs

Education
BS, Computer Science, University of Washington (2018)

Experience
Data Engineer, Tidewell Data Systems (2018-2022)
Backtest Infrastructure Analyst, Brackenridge Systematic (2022-2025)

Kenji Morales is an AI-generated fictional persona responsible for market data engineering and backtest integrity. A computer scientist by training, his invented career began building data pipelines at a fictional analytics firm before moving into systematic research infrastructure. He owns price and dividend-adjusted data sourcing, reproducible backtest code, and checks for look-ahead bias, survivorship issues, and realistic rebalance timing. His distinctive strength is meticulous version control, ensuring every reported result can be regenerated exactly from documented data and code.

AI-generated fictional persona

Elena Vasquez-Moreau

Trading and Investor Communications Analyst
Focus: Trading, execution and investor communications

Certifications: CMT | **Experience:** 6 yrs

Education
BBA, Finance, Georgetown University (2019)
MS, Management, Boston College (2020)

Experience
Execution Associate, Fairhaven Trading Group (2020-2023)
Client Strategy Analyst, Copperline Wealth Partners (2023-2025)

Elena Vasquez-Moreau is an AI-generated fictional persona who handles trading, execution, and investor communications. Her invented career combines ETF execution work at a fictional trading firm with client strategy experience at a fictional wealth adviser. She owns rebalance execution planning, transaction cost and slippage assumptions, and the materials explaining how and why the strategies switch holdings. Her distinctive strength is clarity: she turns systematic rules into straightforward explanations investors can follow, including honest discussion of the strategies' limitations.

AI-generated fictional persona

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Supporting Data: Calendar Years and Rolling Returns

Year	Strategy	SPY	60/40
2007	-0.7%	5.1%	7.7%
2008	-6.8%	-36.8%	-16.7%
2009	14.7%	26.4%	13.3%
2010	-1.8%	15.1%	13.4%
2011	-8.3%	1.9%	8.3%
2012	8.0%	16.0%	11.3%
2013	17.4%	32.3%	15.6%
2014	12.0%	13.5%	12.0%
2015	-4.5%	1.2%	1.8%
2016	13.6%	12.0%	7.8%
2017	21.7%	21.7%	13.8%
2018	1.6%	-4.6%	-2.0%
2019	16.2%	31.2%	21.8%
2020	12.9%	18.3%	16.9%
2021	28.7%	28.7%	15.1%
2022	-22.7%	-18.2%	-16.4%
2023	18.3%	26.2%	17.0%
2024	24.9%	24.9%	14.2%
2025	10.8%	17.7%	14.3%
2026 YTD	9.1%	12.8%	6.3%

Rolling 1-year returns	Strategy	SPY
Best 1-year period	44.6%	77.5%
Worst 1-year period	-23.1%	-47.4%
Periods beating SPY	18.7%	
Median excess vs SPY	-4.1%	
Growth of $10,000 (since walk-forward start)		
Strategy	$42,495	
SPY	$77,253	
60/40	$48,905	

Calendar-year returns are total returns. The final year is year-to-date through 2026-10-16. Rolling figures cover every 252-trading-day window since the walk-forward start. 60/40 is rebalanced daily, without costs.

Important Disclosures

Hypothetical performance. All performance shown is hypothetical and backtested. It does not represent actual trading or actual client accounts. Backtested results are prepared with the benefit of hindsight and have inherent limitations. No representation is made that any account will or is likely to achieve similar results.

Methodology. The strategy holds either SPY or IEF, based on whether SPY closed above a moving average the previous day. The moving average (5, 20, 50 or 200 days) is chosen each January using only data available before that date (walk-forward). Results include a modeled trading cost of 5 basis points per full switch between SPY and IEF. They exclude management fees, taxes and market impact beyond that assumption, and use adjusted closing prices, which include dividends. Partial calendar years are year-to-date.

Momentum methodology. Four monthly momentum sub-strategies (3, 6, 12 and 24-month lookbacks) rank SPY, QQQ, TLT, GLD, IWM and IEF by trailing return at each month-end, and hold 100% of the top performer for the following month, from January 2011. If all six trailing returns are negative, that sub-strategy holds cash at an assumed 0% return for three months and then resumes. Momentum results **do not include trading costs**, while the moving-average strategy is shown net of modeled costs, so comparisons between them favor momentum by the cost of its switches. Cash earning 0% understates what T-bills paid in some periods.

In-sample results. Comparisons of individual moving-average rules over the full period are in-sample. They are shown for context only, and overstate what an investor could have known in advance.

Risk. Investing involves risk, including loss of principal. Trend-following signals can lag reversals, cause frequent switching in choppy markets, and underperform buy-and-hold for long periods. Treasury ETFs carry interest-rate risk and can fall alongside equities.

Data. Prices from Yahoo Finance via the yfinance library. The data has not been independently verified. The risk-free rate is the 13-week Treasury bill yield (^IRX).

Investment team. The investment team members in this presentation are AI-generated fictional personas created by an AI system for illustration. They are not real people, and their stated employers are fictional.

Not advice. This presentation was produced by an automated multi-agent research prototype. It is for informational purposes only, and is not investment advice, an offer, or a solicitation to buy or sell any security.